AIS – AURELIA INDUSTRIAL SYSTEMS

TROUBLESHOOTING GUIDE

AIS Troubleshooting Guide
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1. Document Purpose

This document provides general troubleshooting guidance for AIS industrial assembly systems.

It is intended to help operators and support personnel identify common machine conditions and determine appropriate next steps.

The guide does not authorize users to perform interventions that require qualified technical personnel.

Safety procedures always take precedence over troubleshooting activities.

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2. General Troubleshooting Procedure

When a machine reports an abnormal condition:

1. Identify the machine and serial number.
2. Record the HMI alarm or error code.
3. Determine whether production is stopped or degraded.
4. Check the applicable user manual.
5. Follow the documented operator-level procedure.
6. Do not bypass safety devices.

7. Record actions already performed.
8. Contact AIS Technical Support if the condition remains unresolved.

When contacting Technical Support, provide the machine operating hours and relevant diagnostic information.

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3. Production Stopped

Symptoms

- machine is not producing;
- production cycle is interrupted;
- HMI indicates an active alarm;
- downstream production is affected.

Recommended Actions

- identify the active alarm;
- verify whether an emergency stop is active;
- check machine status on the HMI;
- review the relevant troubleshooting procedure;
- verify whether the condition can be resolved at operator level.

Do not reset alarms repeatedly without understanding the underlying condition.

If production cannot be safely restored, contact Technical Support.

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4. AIS-200 Error E-204

Error Description

E-204: Cooling-system temperature above configured threshold

The error indicates that the cooling-system temperature has exceeded the configured threshold.

Recommended Operator Procedure

1. Stop the machine safely.

2. Do not continue production while the condition remains active.
3. Check the cooling-system status on the HMI.
4. Check coolant level where applicable.
5. Check for visible leakage.
6. Check for abnormal conditions around the cooling equipment.
7. Verify the cooling unit status.
8. Follow the approved recovery procedure.
9. If E-204 persists, contact AIS Technical Support.

Prohibited Actions

Operators must not:

- bypass temperature protection;
- disable the E-204 alarm;
- change the configured temperature threshold;
- continue production solely to avoid downtime;
- open equipment requiring qualified intervention.

Information to Provide Support

When contacting Technical Support for E-204, provide:

- machine serial number;
- current operating hours;
- time of occurrence;
- machine operating state;
- HMI alarm information;
- cooling-system observations;
- recent maintenance activity;
- production impact.

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5. Machine Does Not Start

Possible Causes

Possible causes include:

- active safety condition;
- emergency stop;
- open safety guard;
- unresolved alarm;
- missing operating condition;
- control-system condition.

Recommended Actions

- check the HMI for active alarms;
- verify emergency-stop status;
- verify that machine guards are correctly closed;
- follow the startup procedure;
- review the applicable machine manual.

Do not bypass safety devices to start the machine.

If the machine remains unable to start, contact Technical Support.

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6. Unexpected Machine Stop

Possible Causes

Possible causes include:

- safety-system activation;
- machine alarm;
- sensor condition;
- control-system condition;

- upstream or downstream equipment issue.

Recommended Actions

Record:

- machine state before the stop;
- HMI alarms;
- error codes;
- production status;
- recent maintenance activity;
- any unusual sound, temperature, or behavior.

Do not repeatedly restart the machine if the cause of the stop is unknown.

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7. Reduced Production Performance

Symptoms

The machine operates but production throughput is lower than expected.

Possible Causes

Potential causes include:

- production recipe;
- product characteristics;
- material availability;
- upstream equipment;
- downstream equipment;
- machine configuration;
- robotic cycle conditions;
- repeated alarms or interruptions.

The nominal capacity of the machines is:

Product	Nominal Maximum Capacity

AIS-100	40 units/min
AIS-200	80 units/min
AIS-500	180 units/min

Nominal capacity should not be interpreted as a guaranteed production rate under every operating condition.

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8. Communication Problems

Symptoms

- machine cannot communicate with an external system;
- production monitoring data is missing;
- OPC UA connection is unavailable;
- MES communication is interrupted.

Recommended Actions

For operator-level diagnosis:

- check the HMI communication status;
- identify the affected interface;
- record the approximate time of failure;
- verify whether the machine itself is operating normally;
- contact the appropriate technical support team.

For AIS-500 systems, determine whether the problem affects the machine itself or the MES integration layer.

Network configuration changes should be performed only by authorized personnel.

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9. Repeated Alarms

Repeated alarms should not be treated as independent events without investigation.

When an alarm occurs repeatedly, record:

- alarm code;
- timestamp;
- machine state;
- production conditions;
- operating hours;
- frequency;
- recent maintenance;
- actions taken after previous occurrences.

Recurring conditions may require escalation to Technical Support or Engineering.

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10. Safety-Related Conditions

If a safety-related condition occurs:

1. Stop operation safely.
2. Follow the machine's safety procedure.
3. Do not bypass or disable the safety system.
4. Restrict access where required.
5. Contact qualified personnel.

The AI assistant must never instruct a user to bypass a safety mechanism.

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11. Troubleshooting and Support Tickets

When an existing support ticket concerns the same machine or error condition, the ticket information should be reviewed before creating duplicate support activity.

Relevant ticket information may include:

- ticket status;
- priority;
- category;
- machine;

- error code;
- previous troubleshooting steps;
- support instructions;
- customer responses.

Customer-specific ticket information must only be accessed through authorized systems and permissions.

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12. AI Assistant Usage

The AI assistant may:

- retrieve troubleshooting documentation;
- explain documented procedures;
- retrieve authorized support-ticket information;
- summarize previously performed troubleshooting steps;
- suggest escalation based on documented support policies.

The AI assistant must not:

- override safety procedures;
- authorize unsafe interventions;
- fabricate troubleshooting procedures;
- access unauthorized customer tickets;
- change ticket priority without authorization;
- close a ticket solely based on its own reasoning.

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13. Document Control

Version	Date	Description
1.0	2025-01-15	Initial troubleshooting guide
1.5	2025-07-10	Added E-204 troubleshooting

2.0	2026-01-20	Current troubleshooting procedures

Current approved version: 2.0